

Deep Learning-Based Detection of Phishing Websites

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Week 3 Progress

Week 3 Objectives

- Perform feature importance analysis
- Select top engineered features
- Build hybrid deep learning model
- Evaluate model performance

Feature Importance

Random Forest feature importance was used to identify the most relevant features.

Top 20 features were selected to reduce model complexity and improve generalization.

Hybrid Model Architecture

The model combines two input types:

- URL text processed using CNN
- Engineered tabular features

Outputs from both branches are concatenated before final classification.

- Character embedding
- 1D convolution
- Global max pooling

CNN captures local patterns such as suspicious keywords or URL structures.

Model Results

- Hybrid Model Accuracy: 93.2%
- Improved over CNN-only model
- Demonstrates benefits of combining feature types

Model Comparison

- Random Forest: 96.9%
- LSTM: 87.3%
- CNN: 91.9%
- Hybrid Model: 93.2%

Conclusion

- Hybrid models improve phishing detection performance
- Engineered features remain highly valuable
- Combining deep learning with classical features is effective